Amitesh Gupta

Doctoral Candidate | Ecosystem–Climate Interactions | IIT Bombay
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Research Profile

PhD researcher studying the dynamic interactions between the terrestrial ecosystem and climate conditions. My focus is to comprehend the widespread and complex patterns in ecosystem response to hydroclimatic dry extremes and elucidate their changing vulnerability from the past to the near future. I explore plant-soil-atmosphere processes using eddy covariance flux data and examine physiological and structural changes induced by rapid, instantaneous, or long-duration stress in the land and atmosphere using satellite observations, reanalysis datasets, and climate model simulations. I study how dry events propagate from the atmosphere to the land and reshapes carbon and water fluxes across biomes at varying timescales.

My research bridges the plant ecophysiology, ecosystem-scale carbon cycle, and information-theoretic attribution methods. I am motivated to simulate Earth's carbon and nitrogen cycles using an optimality-based dynamic global vegetation model to understand the impacts of extreme perturbations in the energy and water cycles driven by changes in the climate system.

Education

PhD, Ecosystem–Climate Interactions (Engineering Sciences)
Indian Institute of Technology Bombay, Mumbai, India *2021 – 2026*
Centre for Studies in Resource Engineering (CSRE)

Thesis: Terrestrial ecosystem response to the propagation of extreme dry events from the atmosphere to the land surface.

- Causal influence of initial land-atmosphere and meteorological drought on propagated soil moisture droughts.
- Contrasting response of terrestrial ecosystem based on distinct remote sensing-based descriptors during concurrent (co-occurrence with meteorological drought) and propagated (from meteorological drought) soil moisture droughts.
- Relative influence of soil moisture and vapor pressure deficit on vegetation water content, ecosystem productivity, canopy greenness, and biomass during different stages of flash drought evolution.
- Attribution of low soil moisture and high air temperature on terrestrial ecosystem's physiology and energy partition during the resistance and recovery from compound drought and heat extreme events.
- Changes in the influence of soil moisture and vapour pressure deficit on terrestrial ecosystem's productivity and water-use efficiency with altering concurrent dryness in land-atmosphere.
- Drivers of net terrestrial ecosystem productivity during varying intensity of compound land-atmosphere dryness.

M.Tech, Remote Sensing and GIS (Atmospheric Science specialisation)
Indian Institute of Remote Sensing, Dehradun, India *2017 – 2019*

- **Dissertation**: Spatiotemporal variability in satellite-derived particulate matter concentration in India

M.Sc, Geography (Climatology specialisation)
Savitribai Phule Pune University, Pune, India *2015 – 2017*

- Gold Medallist **| Dissertation**: Temperature trend analysis over major Indian cities

Research Experience

Carbon Cycling: Ecosystem productivity (GPP, NEP), and energy partitioning (T/ET) and coupling among carbon-water-energy fluxes under drought stress using eddy covariance observations (FLUXNET, SAPFLUXNET); ecosystem resistance and resilience using remote sensing-based vegetation descriptors (GPP, LAI, SIF, VOD, NIRv).

Hydro-climatology: Drought propagation dynamics; compound drought and heatwaves; flash drought; wildfires; long-term climate variabilities using reanalysis and climate model simulations (ERA5, GLEAM, CRU, GISTEMP, CMIP6 etc.).

Statistical Methods: Information theory (normalised conditional transfer Shannon entropy; Rényi and KSG estimators of conditional mutual information); Partial Information Decomposition (BROJA-PID); non-linear (KDE) partial correlation; Copula transformation; Average Marginal Effect; multivariate linear and non-linear (MLP-based) Granger causality.

Plant Ecophysiology: Abiotic stress (effect of combined or individual hot/dry condition); legacy effect; phenological shifts (early fluorescence, leaf senescence); carbon allocation and water-uptake, stomatal conductance theory (FvCB).

Professional Experience

GIS Analyst

GIS Cell, Dehradun, India *2020 – 2021*

Responsibilities: Resource mapping, Geodatabase preparation, Monitoring disaster impact (landslide, wildfire, cloudburst)

Technical Skills

Programming: Python (primary language); R (intermediate), Julia.

Procedural programming: Modular Pipeline, multi-core parallelisation (using Dask, Delayed) for big-data analysis.

Object-oriented programming: Encapsulation (use of Xarray, Rasterio), Inheritance.

Python Libraries (specific): Pyarrow, Statsmodels, Scikit-learn, SciPy, shutil, json, GeoPandas, Optuna, Keras, PyTorch.

Modelling frameworks: Machine Learning, Time-Series Analysis, Numerical Weather Prediction (WRF).

Geospatial software: QGIS, ArcGIS, SNAP, PolSARPro, GrADS, Panoply, ERDAS Imagine, ENVI.

Publications

Total citations: 1149 (July 2026) | h-index: 15 | i10-index: 16

Selected publications (full list in *Google Scholar*)**:**

- **Gupta A.**, Lanka K., Mishra A., Wang L.: Dominant Role of Soil Moisture Regulating Vegetation Water Content During Flash Drought Evolution. *Geophysical Research Letters* (2026). doi.org/10.1029/2025GL119366
- **Gupta A.**, Lanka K.: Role of initial conditions and meteorological drought in soil moisture drought propagation: An event-based causal analysis over South Asia. *Earth's Future* (2024). doi:10.1029/2024EF004674
- **Gupta A.**, Saha S., Lanka K.: Intensifying compound dryness amplifies soil moisture control on water-use efficiency and vapor pressure deficit control on productivity across global terrestrial ecosystems. *Environmental Research Letters* (In Press)
- **Gupta A.**, Lanka L.: Impact of Climate trends on Ecosystem Productivity. *IEEE-IGARSS* 2025, Brisbane, Australia. doi:10.1109/IGARSS55030.2025.11243994

Additional peer-reviewed publications: 12 more articles in Q1/Q2 journals, including *Atmospheric Research*, *Geoscience Frontiers*, *Geomatics, Natural Hazards and Risk*, *Atmospheric Pollution Research*.

In Preparation & Under Review:

- Contrasting resistance and resilience of terrestrial ecosystems during concurrent and propagating droughts.
- Shifts in the importance of land-atmospheric drivers to net ecosystem productivity with increasing compound dryness in land-atmosphere: Lessons from global-scale analysis using FLUXNET observations.
- Attribution of low soil moisture and high air temperature to physiological state and energy partitioning of terrestrial ecosystems during compound drought and heat extreme events.
- Shifting role of climate drivers on global greening-browning.

Review experience: Reviewed more than 90 manuscripts for more than 30 journals, including *Nature Communication*, *Geophysical Research Letters*, *AGU Advances*, *Earth's Future, Journal of Environmental Management*, *Journal of Arid Environment*, *Remote Sensing Applications: Society and Environment*, *International Journal of Disaster Risk Reduction*, *Hydrological Sciences*, *Science of the Total Environment*, *Water*, *Remote Sensing*, *Earth systems and environment* etc.

Awards & Grants

- ECS Travel Support, *EGU* 2026, Vienna, Austria
- IEEE Travel Grant, *IGARSS* 2025, Brisbane, Australia
- Gold Medallist, M.Sc. Geography - Savitribai Phule Pune University, Pune.

Supervision, Collaboration & Conference Presentation

- Supervision of master's students for individual dissertation at CSRE, IIT Bombay.
- Presentation of research works at international conferences (EGU 2025, 2026; IGARSS 2025; ISPRS 2019).
- Active global research collaborations (India, US, Belgium, France, Germany, China, Austria, Switzerland, Australia).